

1 Route

Current graph G_t

4 profile scores
+ 4 graph statistics

Learned router

$p(\text{repair}) + \text{scope prior } \pi$

Hard violations or
supplied candidates
override a low trigger.

Utility $U(a)$

Quality gain + hard-violation reduction – edit cost
+ log scope prior + proposal confidence

2 Propose

Model proposals

Fixed edit bundles
grouped by relation

Rule proposals

Detected violations
+ explicit source fields

Delete

Retype

Add

Scope prior π

Deduplicate candidates

3 Evaluate trials

Independent copies of G_t

$a_1 \rightarrow$  G_a

$a_2 \rightarrow$  G_a

$a_3 \rightarrow$  G_a

Recompute profile and flags
Check bounds; calculate $U(a)$

Trial-state constraints

Restoration bounds · bounded component regression
Density bound · nonempty-graph guard

4 Select

Feasible?

$U(a) > 0?$

Highest utility
among eligible edits

Commit one edit

$G_t \rightarrow G_{t+1}$

Reassess the committed graph; repeat within the 12-step horizon.

Stop if no eligible edit exists or the 12-step horizon is reached.

Early exit: no supplied candidates, and either a low trigger without hard flags or no detected violations.